

Feasibility Memo: AI-Assisted ED Triage Support System

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Project Title: Explainable AI-Assisted Emergency Department Triage Support System for Resource Constrained Healthcare Settings

Verdict

Proceed to a baseline triage model, with caveats. The Yale EMMLC triage extract provides a real triage label, complete structured and complaint data, and no physiologically impossible values. There are sufficient grounds to build and test an explainable baseline model, provided the noted limits (representativeness, single-snapshot structure, demographic imbalance) are respected and monitored throughout the pilot.

Dataset Summary

The dataset contains 55,121 de-identified adult ED encounters (ages 18–107, mean 55.3) across 225 features: 25 structured fields (demographics, vitals, arrival context, disposition) and 200 sparse chief-complaint flags (cc_*), each a 0/1 indicator.

The target, esi (Emergency Severity Index), has 5 ordinal levels, from most urgent (1) to least (5). The distribution is realistically skewed toward mid-acuity: ESI 3 (49.0%), ESI 2 (32.5%), ESI 4 (16.1%), ESI 5 (2.2%), ESI 1 (0.1%). Vital signs are recorded at triage (heart rate, systolic/diastolic BP, respiratory rate, oxygen saturation, temperature, glucose); temperature is in Fahrenheit, not Celsius. Most patients (86.9%) present with a single chief complaint, keeping the input pattern per patient tractable and explainable rather than an unmanageable 200-column sprawl.

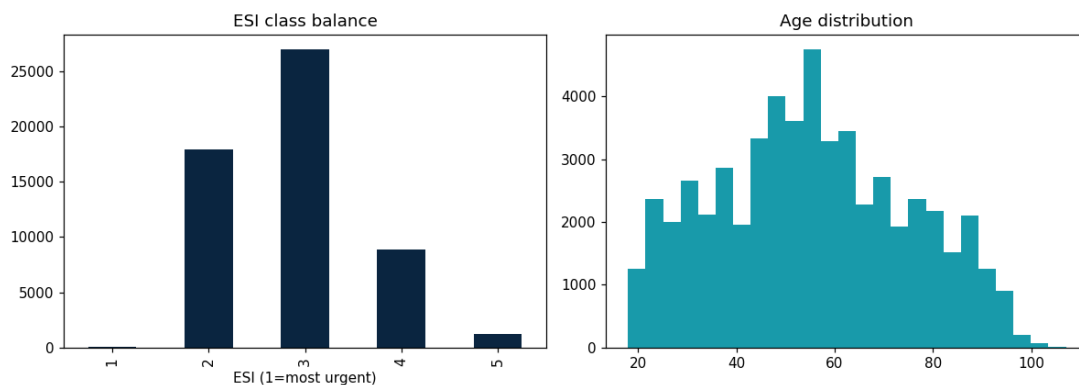


Figure 1: ESI class balance (left) and age distribution (right).

Top 3 Concerns

- Outcome leakage

Disposition (Admit/Discharge) and previousdispo are known only after the triage decision. Including either as a model input would leak future information into a triage-time prediction. Both are therefore excluded by design.

- Demographic imbalance and potential bias

ESI distribution within race category (Figure 2) is not uniform. For example, the Native Hawaiian/Pacific Islander group shows the widest spread across ESI levels, but this group numbers only 20 patients in the entire dataset, making the pattern statistically unreliable rather than a confirmed disparity. This must be

monitored, not concluded and is exactly the kind of signal the pilot's bias-monitoring commitment needs to track.

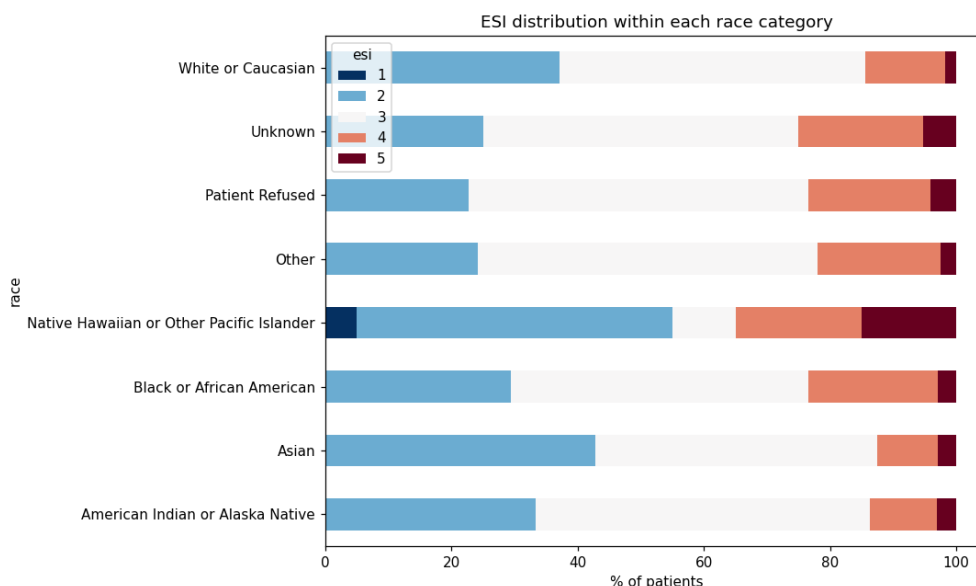


Figure 2: ESI distribution within race category (Native Hawaiian/Pacific Islander, $n=20$, too small to draw conclusions from).

- Representativeness for a Caribbean ED

This is a single US academic-hospital sample. Case-mix, demographics, insurance/access patterns, and arrival modes will differ from a Caribbean ED context (e.g., Mercer General) and this cannot be resolved with this dataset alone, it must be tested during the pilot itself.

Top 3 Reasons to Proceed

- A real, complete triage label

ESI is present for 100% of encounters- a genuine target to learn from, not a synthetic stand-in.

- Usable, clinically coherent vitals

All 7 vital-sign fields show zero missing values and zero clinically impossible readings (per hard physiological bounds). Statistical outliers exist (e.g. 3,554 in temperature, 5,648 in glucose) but represent real extreme presentations, not data errors and vitals separate acuity in the expected clinical direction: oxygen saturation correlates most strongly with ESI ($r = 0.18$) among single vitals, and systolic/diastolic BP show the expected internal correlation ($r = 0.63$).

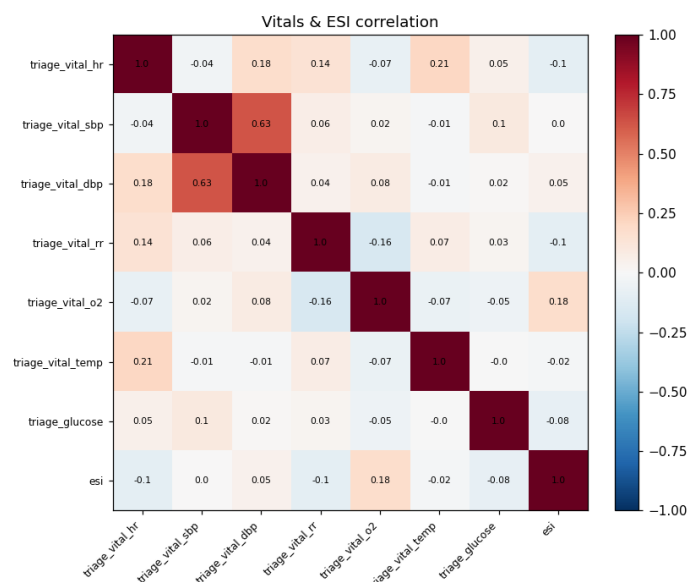


Figure 3: Correlation heatmap of vitals and ESI.

- A documented, reproducible pipeline

Every step, including data loading, missing data assessment, outlier detection, correlation analysis, and demographic breakdowns, is documented in a version-controlled notebook, allowing the analysis and its underlying assumptions to be independently audited.

Caveats

- Temperature is recorded in °F, not °C. Any clinical threshold logic must account for this explicitly.
- Chief-complaint flags are sparse; the majority carry very low prevalence and offer limited individual signal.
- Correlations reported here are associational, not causal, and not a measure of model performance, they indicate what's worth testing, not what's proven.
- This dataset is a single triage-time snapshot per patient, not a time-series of readings, it can support proof-of-concept correlation analysis but cannot, on its own, validate the "earlier identification of deteriorating patients" goal, which needs sequential/repeated-measures data.
- External validity for a Caribbean ED deployment is untested and should be a named pilot objective, not an assumption.

Preliminary Feature Shortlist (Hypothesis- Worth Testing)

Ranked by correlation strength with ESI (leakage columns excluded), combined with clinical intuition:

- 1. triage_vital_o2
- 2. triage_vital_rr
- 3. triage_vital_hr
- 4. triage_glucose
- 5. cc_chestpain
- 6. cc_shortnessofbreath
- 7. age
- 8. cc_abdominalpain
- 9. triage_vital_temp
- 10. Arrivalmode